Offline Channel Assumption Engine Flow and Logic

How the financial-model agent turns events, partnerships and BD conversations into lead and deal assumptions, by industry and country, for 29 industries and 21 MENA countries

FUNNEL MATH | COUNTRY AND INDUSTRY FACTORS | AGENT FLOW | WORKED EXAMPLES

All base rates and factor ratings are expert PRIORS (v0.1), not market statistics.

They are meant to be reviewed by your regional BD team and replaced with CRM actuals.

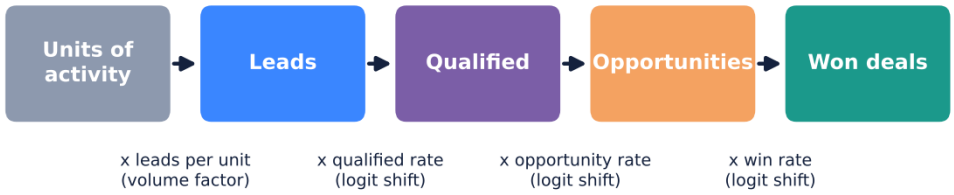
Version 0.1 | 14 November 2026 | Code and full 14,000-row matrix: assumption_engine/ in the repository

SECTION 1

What this is and how the agent uses it

Your financial-model agent needs numbers for channels that CRMs capture poorly: **events, partnerships and business-development conversations**. This document defines one repeatable logic that turns *(country, industry, deal type, channel)* into funnel assumptions: leads per activity, qualified, opportunity and win rates, sales-cycle time, partner ramp-up, a P10 to P90 range, and the reasons behind each number.

The same logic works for all **21 countries** and **29 industries** in your two files, so Saudi Arabia B2B e-commerce partnerships get different numbers from Egypt without anyone writing 609 separate rules. Each country and industry is described by a few rated factors; each channel says which factors matter for which funnel stage.



Time lag: deals land after the cycle time (days), scaled by country, industry and season. Revenue = deals x ACV (ACV comes from your model).

Figure 1. The funnel chain. Each arrow is a rate that country and industry factors shift.

Read this before using any number

I did **not** take the base rates or the 1 to 5 factor ratings from a dataset. They are **my structured starting estimates (priors)** so the agent has a working, explainable default. They must be reviewed by people who sell in these markets and replaced by your CRM actuals as they accumulate. Every output is labelled **prior**, **blended** or **company actuals** so nobody mistakes a prior for evidence.

What you get

- This PDF: the flow, the math, the factor dictionary, all ratings, worked examples and the agent contract.
- **assumption_engine/** in the repository: the engine (**engine.py**, Python standard library only), editable JSON config, 11 passing tests, and the full base matrix **data/matrix_base.csv** (14,000 rows: every valid country, industry, deal type and channel).
- Both your uploaded lists are copied into **data/** and every name in them is tested to resolve (for example 'Uae' resolves to UAE).



Figure 2. The agent flow, step by step. Steps 3 and 9 are where a person stays in control.

SECTION 2

The logic, in plain words and formulas

Every channel has a base funnel for a neutral market (all factor ratings at 3, B2B enterprise). Country and industry factors then push each stage up or down. Ratings run 1 to 5; the engine uses the distance from neutral, so a rating of 5 is +2 and a rating of 1 is -2.

Item	Formula	Why
Leads per unit of activity	base x deal-type multiplier x exp(sum of coefficient x factor distance)	Multiplicative, so effects stack and can never go negative.
Qualified, opportunity, win rate	sigmoid(logit(base rate) + deal-type shift + sum of coefficient x factor distance)	Logit shifts keep every rate strictly between 0% and 100% however extreme the market.
Sales-cycle days	base x deal-type multiplier x exp(sum of coefficient x factor distance)	Slow buyers, regulation and big committees stretch time.
Deals per unit	leads x qualified x opportunity x win	The number the financial model needs.
Range (P10 to P90)	4,000 simulated draws; each stage rate and volume varies with a spread set by country tier	Honest uncertainty: wider for tier C and D countries.
Blend with actuals	(prior x 15 + successes) / (15 + trials) for rates; similar for volume and cycle	15 pseudo-observations: 15 real trials equal the prior. More data, more weight, narrower range.
Monthly forecast	units x leads x rates, then deals land over the cycle time (5 lag points, split across months)	Revenue follows deals by the sales-cycle delay, not the same month.

Deal types, and where they apply

Deal type	Leads x	Qualified shift	Opportunity shift	Win shift	Cycle x	Channels modelled
B2B_SMB	1.3	+0.1	+0.1	+0.2	0.6	all 7
B2B_ENT	1	+0	+0	+0	1	all 7
B2G	0.7	+0	-0.1	-0.3	1.8	all 7
B2C	3	+0.3	+0.8	-0.5	0.25	events, referral, reseller

B2G also adds a volume effect from how much of demand the state drives (+0.12 per step of the country's 'state share of demand' factor). Shifts are log-odds, not percentage points.

Gates: when the engine refuses or warns

Gate	Countries (prior)	What the agent does
normal	Saudi Arabia, UAE, Qatar, Kuwait, Bahrain, Oman, Egypt, Jordan, Morocco, Tunisia, Turkey	Produces assumptions.
elevated	Lebanon, Algeria, Iraq, Palestine	Produces assumptions with a warning and wide ranges; asks the local team to confirm.
review_required	Libya, Iran, Syria, Sudan, Yemen	**Returns no numbers.** Needs compliance and legal review first (sanctions, conflict or access constraints). Current status must be checked by your team; the gate is a prior, not legal advice.
not applicable	Deal type and channel pairs not modelled (for example B2C with cold BD meetings)	Says so instead of inventing.

SECTION 3

Factor dictionary: how to rate a country or an industry

Nine country factors and six industry factors. Anyone on your team can re-rate a cell with these anchors; the engine then recomputes everything. 3 is neutral.

Country factor	1 means	5 means	Mainly moves
Relationship dependence	Transactional, contract-first buying	Deals need personal trust and introductions	Warm channels up, cold BD down
Digital B2B procurement	Paper and phone procurement	E-procurement and e-signature are normal	Fewer events needed; faster cycle
Buyer decision speed	Approvals take very long	Decisions in days or weeks	Cycle time, opportunity rate
Event ecosystem	Few relevant events	Dense calendar of major regional events	Event leads and qualification
Partner ecosystem	Few credible distributors or integrators	Deep partner and integrator base	Partnership leads and wins
Reachable buyer density	Few reachable buyers	Large reachable buyer base	Lead volume
State share of demand	Private-sector led	State drives most demand	B2G volume, cycle length
Payment, contract, FX friction	Smooth payment and enforcement	Long delays, FX or enforcement problems	Win rate down
Local-language need	English is enough	Local language near-mandatory	Reserved for later use; recorded, not yet in formulas

Industry factor	1 means	5 means	Mainly moves
Sales-cycle length	Days to weeks	Nine months or more	Cycle time
Ticket size	Small deals	Very large deals	Qualification, event ROI
Partner dependence	Direct sales	Partner-led	Partnership volume and wins
Event dependence	Buyers rarely buy via events	Events are the main discovery route	Event leads
Regulation / approvals	None	Licences and approvals needed	Win rate down, cycle up
Buying-committee size	One decision maker	Large committee	Opportunity and win down, cycle up

Local-language factor

The nine country factors include local-language need, recorded for each country, but the v0.1 formulas do not use it yet. It is there so you can decide how it should affect each channel, rather than my guessing.

SECTION 4

Country ratings (priors for your team to review)

Country	Tier	Gate	Rel	Dig	Spd	Evt	Par	Mkt	Gov	Fric	Loc	Ramadan sens.
Saudi Arabia	A	normal	5	3	3	5	4	5	5	2	4	1
UAE	A	normal	4	5	4	5	5	4	3	1	2	1
Qatar	A	normal	5	4	3	3	3	2	5	2	3	1
Kuwait	A	normal	5	3	2	2	3	2	5	3	4	1
Bahrain	A	normal	4	4	3	3	3	1	4	2	3	1
Oman	A	normal	5	3	2	2	2	2	5	3	4	1
Egypt	B	normal	4	3	2	4	4	5	4	4	5	0.9
Jordan	B	normal	4	3	3	3	3	2	4	3	4	0.9
Morocco	B	normal	4	3	2	3	3	3	4	3	4	0.9
Tunisia	B	normal	4	3	2	2	2	2	4	4	4	0.9
Turkey	B	normal	3	4	3	4	4	5	3	4	5	0.5
Lebanon	C	elevated	4	2	2	2	2	1	3	5	3	0.5
Algeria	C	elevated	4	2	1	2	2	3	5	5	5	0.9
Iraq	C	elevated	5	2	2	2	2	3	5	5	5	0.9
Palestine	C	elevated	4	2	2	1	2	1	3	5	4	0.8
Libya	D	review_required	5	1	1	1	1	2	5	5	5	0.9
Iran	D	review_required	4	2	2	3	3	4	5	5	5	0.9
Syria	D	review_required	5	1	1	1	1	2	5	5	5	0.9
Sudan	D	review_required	5	1	1	1	1	2	5	5	5	0.9
Yemen	D	review_required	5	1	1	1	1	1	5	5	5	0.9

Columns: Rel relationship dependence, Dig digital procurement, Spd decision speed, Evt event ecosystem, Par partner ecosystem, Mkt buyer density, Gov state share of demand, Fric friction (higher is worse), Loc local-language need. Tier sets the width of the range: A 0.25, B 0.35, C 0.45, D 0.60 (spread parameter). 'Uae' in your file is read as UAE; Turkey is kept as written in your list. Shaded rows are gated.

The 29 industries in your list

Industry	Cycle	Ticket	Partner	Event	Reg.	Cmte
Manufacturing	4	4	4	5	3	4
CloudTech	4	4	5	4	3	4
Cyber Security	4	4	4	4	4	4
Entertainment	3	3	3	4	3	2
AutoTech	5	4	4	4	4	4
Logistics	3	3	4	4	3	3
InvestTech	4	4	3	3	5	3
Data and Analytics	4	3	4	3	3	4
EdTech	4	3	3	3	5	4
MediaTech	3	2	3	4	3	2
E-Commerce	3	3	4	4	2	3
Gaming	3	2	3	4	3	2
FinTech	5	4	4	4	5	4
Telecommunications	5	5	4	3	5	5
SaaS	3	3	4	4	2	3
TravelTech	3	3	4	5	3	3
AdTech	2	2	3	4	2	2
Blockchain	4	3	3	4	5	3
AgriTech	4	3	4	4	4	3
Hardware	4	4	5	5	3	3
Biotechnology	5	5	4	4	5	5

Industry	Cycle	Ticket	Partner	Event	Reg.	Cmte
HealthTech	5	4	4	4	5	5
Apparel	2	2	4	5	2	2
AI	4	4	4	5	4	4
PropTech & ConTech	4	4	4	5	4	4
SportTech	3	3	3	4	3	3
Sustainability	4	4	4	4	5	4
FoodTech	3	3	4	5	4	3

SECTION 5

Channels: base funnels and which factors move which stage

Base funnels are for a neutral market, B2B enterprise. 'Unit' is what the financial model counts (one event, one active partner for a month, one meeting held).

Channel	Unit	Leads / unit	Qualified	Opp.	Win	Cycle (days)	Ramp (mo)	Deals / unit
EVT_EXHIBIT Event: exhibition / booth	event_participation	60	25%	30%	20%	120	0	0.900
EVT_HOSTED Event: hosted roundtable / workshop	hosted_event	25	40%	35%	25%	90	0	0.875
PAR_RESELLER Partnership: reseller / distributor	active_partner_month	2.5	50%	40%	25%	150	4	0.125
PAR_STRATEGIC Partnership: strategic / co-sell / integration	active_partner_month	1.7	60%	45%	30%	180	6	0.138
PAR_REFERRAL Partnership: referral	active_partner_month	1	70%	50%	35%	75	2	0.122
BD_COLD BD conversation: cold / outbound meeting	meeting_held	1	20%	35%	15%	150	0	0.010
BD_WARM BD conversation: warm intro / network meeting	meeting_held	1	45%	45%	30%	100	0	0.061

Priors only. Event leads are contacts captured; a 'lead' for a BD meeting is the prospect met. Define these once in your CRM so actuals match the units.

Factor sensitivities by channel (coefficient per rating step)

Channel	Leads (V)	Qualified (Q)	Opportunity (O)	Win (W)	Cycle (T)
EVT_EXHIBIT	event ecosys. +0.10 buyer density +0.08 event dep. +0.10 digital -0.03	event ecosys. +0.05 event dep. +0.08 ticket -0.04	relationship +0.04 speed +0.04 committee -0.05	friction -0.08 relationship +0.05 regulation -0.06 committee -0.05	cycle length +0.12 speed -0.10 state demand +0.05 regulation +0.08 committee +0.06
EVT_HOSTED	relationship +0.06 buyer density +0.04 event dep. +0.05	relationship +0.10 ticket +0.03	relationship +0.08 speed +0.04	relationship +0.08 friction -0.08 regulation -0.05 committee -0.04	cycle length +0.10 speed -0.10 regulation +0.06
PAR_RESELLER	partner ecosys. +0.15 buyer density +0.08 partner dep. +0.12	partner ecosys. +0.06 partner dep. +0.06	partner ecosys. +0.05 relationship +0.04	partner ecosys. +0.06 friction -0.07 regulation -0.05	cycle length +0.12 partner ecosys. -0.06 speed -0.08 regulation +0.07
PAR_STRATEGIC	partner ecosys. +0.10 partner dep. +0.10 state demand +0.04	partner ecosys. +0.05 ticket +0.04	partner ecosys. +0.05 relationship +0.06 committee -0.06	relationship +0.06 friction -0.07 regulation -0.06 committee -0.05	cycle length +0.14 committee +0.08 speed -0.08
PAR_REFERRAL	partner ecosys. +0.08 relationship +0.08 partner dep. +0.05	relationship +0.08	relationship +0.08	relationship +0.08 friction -0.07 regulation -0.04	cycle length +0.08 speed -0.08

Channel	Leads (V)	Qualified (Q)	Opportunity (O)	Win (W)	Cycle (T)
BD_COLD	buyer density +0.06 relationship -0.06	relationship -0.10 digital +0.04	relationship -0.08 speed +0.05	relationship -0.04 friction -0.07 regulation -0.05 committee -0.05	cycle length +0.12 speed -0.10 relationship +0.06
BD_WARM	relationship +0.06 buyer density +0.05	relationship +0.10	relationship +0.08 speed +0.05	relationship +0.08 friction -0.07 regulation -0.05 committee -0.04	cycle length +0.10 speed -0.08 state demand +0.04

Example reading: for BD_COLD qualified rate, relationship -0.10 means each step of relationship dependence above neutral lowers the log-odds of qualifying a cold meeting by 0.10. These coefficients are the main thing to calibrate once you have data (Section 9).

SECTION 6

Results: how the same channel differs by country

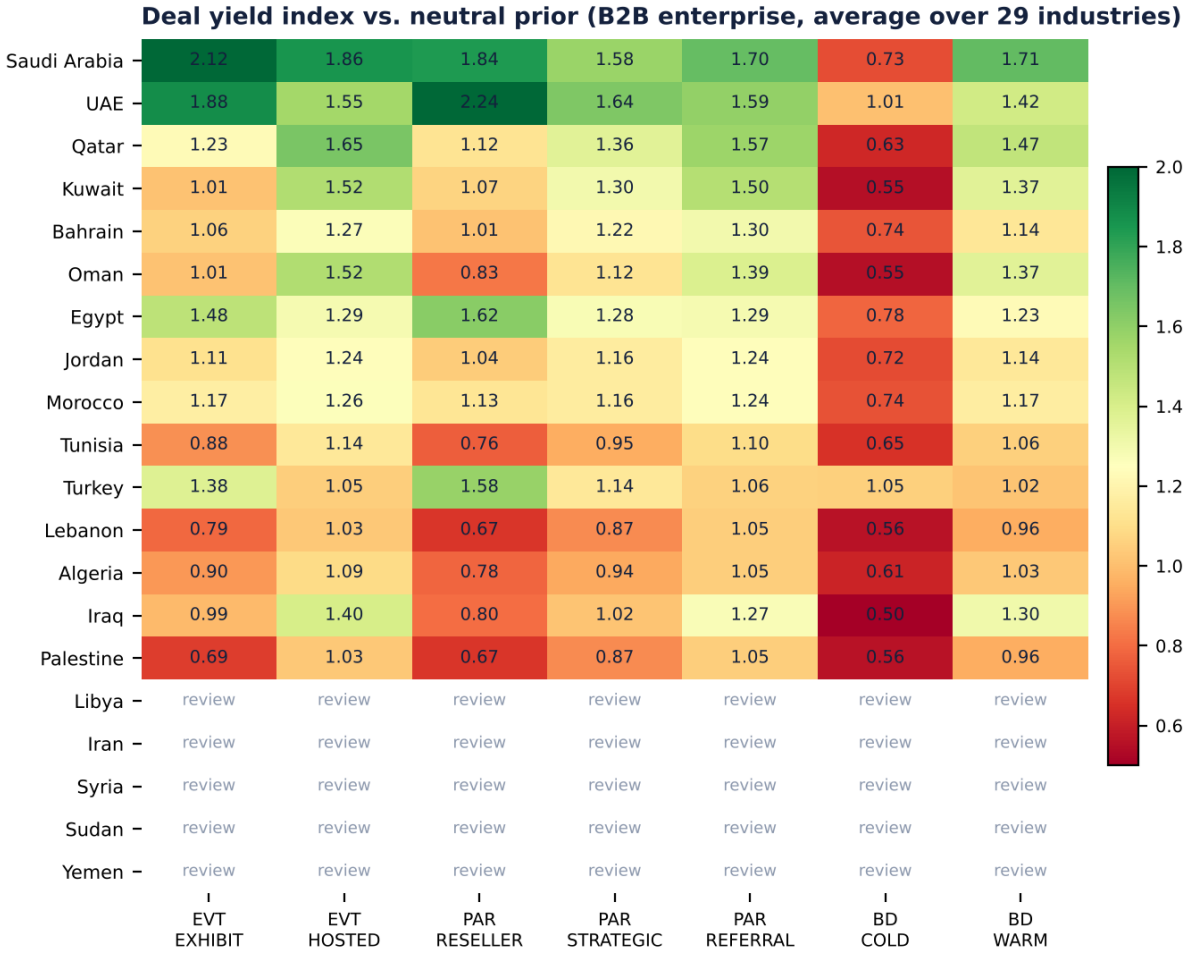


Figure 3. Deal yield per unit of activity relative to the neutral prior, averaged over all 29 industries (B2B enterprise). Green means better than neutral. Cold BD is weak in relationship-driven markets; partnerships and events are strongest in Saudi Arabia, UAE and Egypt on these priors. Gated countries show 'review'.

Worked example: E-Commerce, B2B enterprise, strategic partnership

Metric (per active partner-month)	Saudi Arabia	UAE	Egypt
Leads	2.25	2.29	2.16
Qualified rate	61%	62%	61%
Opportunity rate	49%	49%	48%
Win rate	35%	36%	31%
Sales cycle (days)	180	166	195
Deals per partner-month	0.241	0.250	0.196
Deals per partner-month, P10 to P90	0.149 to 0.368	0.156 to 0.383	0.100 to 0.349
Partners needed for 12 deals a year (base)	4.2	4.0	5.1

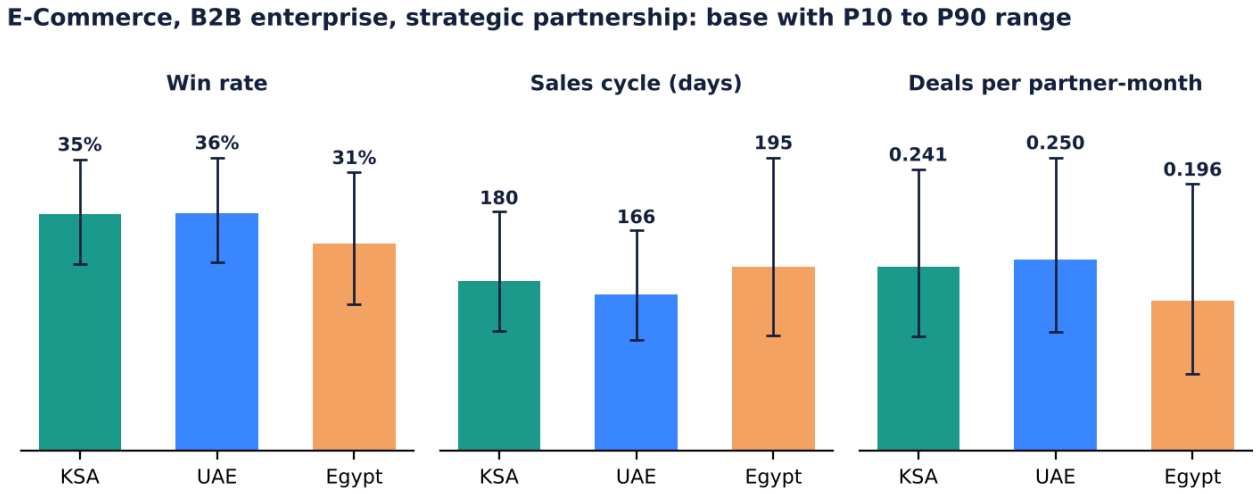


Figure 4. Base value with P10 to P90 bars. Ranges overlap: the engine says KSA and UAE look better than Egypt on these priors, but it is not certain.

Why the numbers differ (top drivers, from the engine)

Metric	Saudi Arabia	Egypt
Win rate	Country: Relationship dependence (rating 5): +0.12 logit Country: Payment / contract / FX friction (rating 2): +0.07 logit Industry: Regulation / approvals exposure (rating 2): +0.06 logit	Country: Payment / contract / FX friction (rating 4): -0.07 logit Country: Relationship dependence (rating 4): +0.06 logit Industry: Regulation / approvals exposure (rating 2): +0.06 logit
Sales cycle		Country: Buyer decision speed (rating 2): +8.3%
Leads	Country: Partner / channel ecosystem maturity (rating 4): +10.5% Industry: Partner dependence (rating 4): +10.5% Country: State share of demand (rating 5): +8.3%	Country: Partner / channel ecosystem maturity (rating 4): +10.5% Industry: Partner dependence (rating 4): +10.5% Country: State share of demand (rating 4): +4.1%

Reading: friction (payment delays, FX, contract enforcement) is the main reason the Egypt win rate and cycle are worse than Saudi Arabia on these priors; relationship strength and partner ecosystem lift Saudi leads. If your regional team disagrees with a rating, change it in config/countries.json and re-run.

SECTION 7

From per-unit assumptions to a monthly forecast

The agent usually receives a **plan**, not a single combination: how many events, partners and meetings each month. The forecast function applies seasonality and partner ramp-up, then lands deals after the sales-cycle delay. Revenue is deals x ACV, using the ACV from your model.

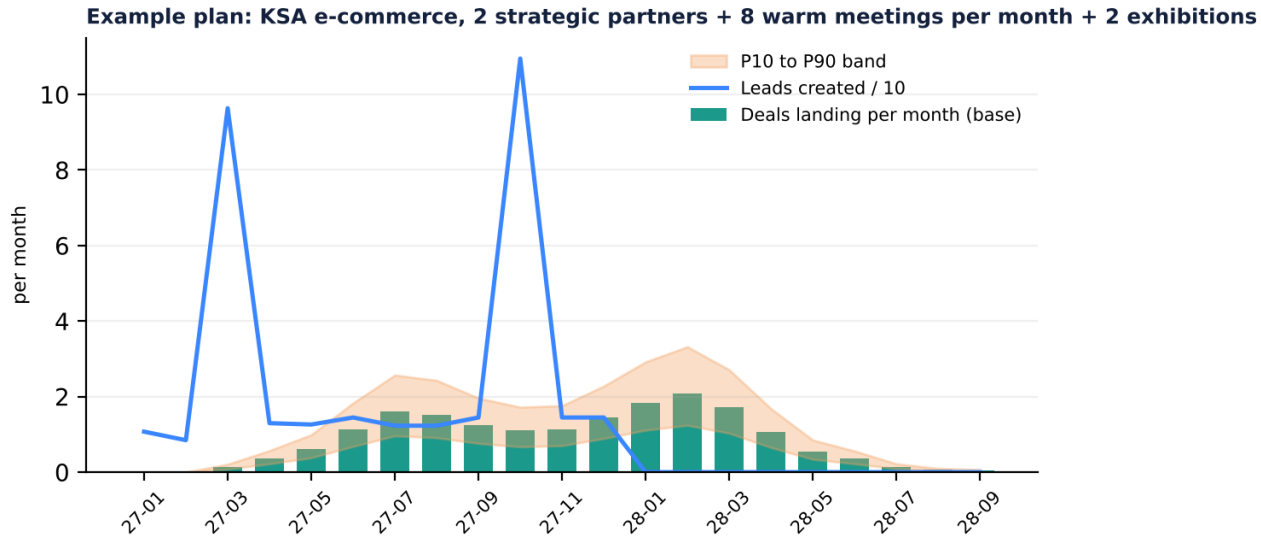


Figure 5. Example plan: KSA e-commerce, two strategic partners from January 2027 (ramping up), 8 warm meetings per month, exhibitions in March and October. Deals trail the lead spikes by the sales cycle; the Ramadan 2027 window (about 8 Feb to 9 Mar) dampens early leads.

Seasonality and ramp rules

Effect	Rule (config/model.json)	Note
Ramadan	Lead volume x0.65, cycle x1.25, scaled by the share of the month inside Ramadan and by the country sensitivity (0.5 to 1.0)	Windows for 2026 to 2028 are approximate expected dates. Confirm with official calendars.
Eid al-Adha	Volume x0.5 for about 5 days	Approximate dates; same caveat.
Summer	July and August: volume x0.85, cycle x1.1	Prior; tune per country from your own calendar.
Partner ramp-up	Productivity rises linearly to 100% over the ramp months (reseller 4, strategic 6, referral 2)	Pass partner_age_months in the plan.
Deal timing	5 lag points (0.6, 0.8, 1.0, 1.2, 1.5 x cycle) with weights 15, 25, 30, 20, 10%	Total deals are conserved; tested.

What the forecast does not include

- **Cost.** Event, partner and travel costs are inputs from your model. The engine returns units per deal, so cost per deal is units per deal x your unit cost.
- **Pricing and ACV.** Taken from your model, not inferred.
- **Cannibalisation** between channels (a partner lead that an event also generated). Start with separate channels and review overlap in your CRM.
- **Currency.** The engine counts deals and leads; convert revenue in your model.

SECTION 8

The agent contract: tools, rules and output

Tool	Inputs	Returns
list_supported()	none	The 21 countries, 29 industries, 4 deal types and 7 channels with aliases.
get_channel_assumptions()	country, industry, deal_type, channel, optional actuals	Status (OK, NOT_APPLICABLE, REVIEW_REQUIRED), all funnel assumptions with P10/base/P90, drivers, provenance, warnings.
build_forecast()	plan rows (country, industry, deal_type, channel, month, units, partner_age_months), optional acv	Monthly leads, qualified, opportunities, deals (base, P10, P90) and revenue.
blend_actuals()	same as above plus CRM counts (successes and trials per stage, leads and units, cycle mean and n)	Updated assumptions, provenance 'blended' or 'company_actuals', narrower range.

Rules for the agent (put these in its system prompt)

- Only use countries, industries and channels the tools support. If the user's wording is unknown, ask; never map silently to a similar one.
- **Never invent a number.** Every assumption written to the model comes from a tool call and carries its provenance.
- If status is REVIEW_REQUIRED, stop, tell the user why, and do not estimate by analogy to another country.
- State the provenance in plain words: 'prior estimate' versus 'based on your CRM data'. Show the range, not only the base.
- Ask for missing inputs: planned volume per channel, ACV, unit costs, any CRM actuals, whether the deal is SMB, enterprise, government or consumer.
- Explain the top drivers in one sentence ('slower cycle in Egypt mainly because of payment friction').
- Flag for human review when: provenance is prior and the channel is a large share of planned pipeline; country gate is elevated; or the user overrides a factor.

Assumption row written to the financial model (example from the engine)


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{
 "key": "ksa.ecommerce.b2b_ent.par_strategic.win_rate",
 "value": 0.355,
 "low_p10": 0.28,
 "high_p90": 0.437,
 "unit": "share of opportunities won",
 "channel": "PAR_STRATEGIC",
 "provenance": "prior",
 "methodology": "assumption_engine v0.1",
 "drivers": [
  "Country: Relationship dependence",
  "Country: Payment / contract / FX friction",
  "Industry: Regulation / approvals exposure"
 ],
 "warnings": [],
 "review_required": false
}
```

SECTION 9

Making the priors real: calibration plan and limits

The engine is built so that every prior can be replaced by evidence without changing the logic. Order of work:

Step	What to do	Result
1	Have 2 to 3 people who sell in these markets review the country and industry ratings (Sections 4 and 5) and the channel base funnels. Change what they disagree with.	Better priors, written down and versioned.
2	Define CRM units once: what counts as an event participation, a hosted event, an active partner-month, a meeting held, a lead, qualified, opportunity and win. Tag every record with channel, country, industry, deal type.	Actuals that match the engine's units.
3	Export past deals and activities. Load as actuals (blend_actuals). Start with the countries and channels that have the most records.	Provenance changes to blended; ranges narrow.
4	At 50 or more trials in a stage, treat as company actuals; keep the prior as a regulariser for thin cells.	Stable numbers for big cells, sensible defaults for small.
5	Quarterly: compare forecast with actual leads and deals by channel and country. Where the gap exceeds the P10 to P90 range repeatedly, adjust coefficients or ratings.	Learning loop; list changes in a version log.
6	Later, fit the coefficients statistically from your pooled data (a multi-level model across countries and industries) once there are enough deals.	Data-driven factor effects.

Known limits of version 0.1

- **Priors, not data.** Base funnels and all 15 factor ratings come from my judgement. The structure is sound; the numbers are placeholders until reviewed.
- **Country and industry effects add up independently.** There are no interaction terms, for example a heavily regulated industry in a state-led market. Add them once data shows they matter.
- **B2C is simplified.** Only events, referral and reseller channels are modelled for consumer deals.
- **Gates are priors, not legal advice.** Sanctions and access rules change; your compliance team must set the current gate for Iran, Syria, Sudan, Yemen, Libya and any elevated country.
- **Seasonality dates are approximate.** Ramadan and Eid dates move each year and depend on moon sighting.
- **Local-language need is recorded but not yet used.** Decide with your team how it should affect each channel.
- **Your two lists are the whole scope.** A country or industry outside them is refused until it is added to the config.

Repository contents

Path	Purpose
assumption_engine/engine.py	Engine: assume(), forecast(), season(), command line for single lookups and the full matrix.
assumption_engine/config/countries.json, industries.json	Factor ratings, tiers, gates, aliases. Edit these to change the logic.
assumption_engine/config/model.json	Channels, base funnels, coefficients, deal types, uncertainty, seasonality.
assumption_engine/data/matrix_base.csv	Base assumptions for every valid combination (14,000 rows) for loading into a model or sheet.
assumption_engine/data/*_list.csv	Your two uploaded lists.
assumption_engine/tests/test_engine.py	11 tests: name matching for every row in your lists, bounded rates, gates, blending, determinism, deal conservation.